



Merit Education Group

What Is a Variable? — Independent Practice

Algebra 1 • Unit 1 • Lesson 1

Standard: 9.3.5.7

Standard: 9.3.5.8

[Name: _____]

[Date: _____] [Class

Period: _____]

Tier 1: Foundational

Review the vocabulary below. Fill in the blank or circle the correct answer for each problem.

1. A _____ is a symbol, usually a letter, that represents an unknown quantity.

2. In the expression $3x + 5$, the number 5 is called a _____ because it does not change.

3. In $7y$, the number 7 is the _____ and y is the _____.

4. Circle the variable(s) in the expression: $4a + 9 - 2b$

5. Circle the coefficient(s) in the expression: $12m + 3n - 8$

Tier 2: Application

Translate each word phrase into an algebraic expression. Use x to represent the unknown number.

6. A number increased by 9

7. Seven less than a number

8. The product of 5 and a number

9. Twice a number, decreased by 4

Tier 3: Extension

Use variables to model each real-world situation. Show all steps and define your variable clearly.

10. The length of a rectangular garden is 3 feet more than twice its width. If w represents the width, write an expression for the length. Then write an expression for the perimeter of the garden.

11. A cell phone plan costs \$25 per month plus \$0.10 per text message. Let t represent the number of text messages sent in a month. Write an expression for the total monthly cost. Then calculate the cost if 120 text messages are sent.
